

Solutions & Scoring Rubrics: Aldehydes & Ketones Mock

Complete Model Answers by Ajay Choudhary Sir

Chapter / Focus Area: Class 12 Organic Chemistry Solutions

Official Model Solutions

Ans 1. Tollen's Test: Propanal gives silver mirror with $[\text{Ag}(\text{NH}_3)_2]^+$, Propanone does not. Alternatively, Iodoform test gives yellow CHI_3 ppt with propanone but not propanal. [2 Marks]

Ans 2. Step 1: CN^- nucleophile attacks electrophilic carbonyl carbon (sp^2 to sp^3 tetrahedral alkoxide). Step 2: Protonation of alkoxide by HCN gives cyanohydrin. [3 Marks]

Ans 3 (a): $\text{C}_6\text{H}_5\text{COCH}_3 + \text{NH}_2\text{NH}_2 \rightarrow \text{Hydrazone} \rightarrow (\text{KOH/Ethylene glycol, heat}) \rightarrow \text{C}_6\text{H}_5\text{CH}_2\text{CH}_3$ (Ethylbenzene). [2 Marks]

Ans 3 (b): $\text{CH}_3\text{CH}_2\text{COOH} + \text{Cl}_2/\text{Red P} \rightarrow \text{CH}_3\text{CH}(\text{Cl})\text{COOH}$ (2-Chloropropanoic acid). [2 Marks]

Ans 3 (c): $\text{C}_6\text{H}_5\text{COCl} + \text{H}_2$ (Pd/BaSO₄, poisoned by quinoline/S) $\rightarrow \text{C}_6\text{H}_5\text{CHO}$ (Benzaldehyde). [2 Marks]

Official Verified Study Resource • Mentored by Ajay Choudhary Sir

Web: <https://pankaj2816.github.io/ajay-choudhary-chemistry/> • Contact: contact@ajaychemistry.com